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Stress-Testing a Distributed Public-Resource Governance Architecture: Adversarial and Behavioral Agent-Based Evidence

Draft v0.4 — 2026-07-07. Companion to the working paper "Distributed Resource Governance: An Architecture for Transparent Public Resource Allocation with Adversarial Validation" (doi:10.5281/zenodo.21228492), hereafter "the architecture paper".

Abstract

The architecture paper proposes Core v0, a public-resource allocation architecture in which citizens direct budget shares to concrete projects under milestone-gated, independently verified disbursement, with a published planning vector absorbing non-participation. This companion paper reports the computational experiment program that stress-tests that architecture beyond its own simulation suite. We contribute a reproducible adversarial agent-based framework (deterministic engines, paired seeds, pre-registered designs with public prediction accounting, and a traceability contract binding every mechanism to its architectural anchor), a method for generating behaviorally realistic participation endogenously — including calibration by large-language-model-elicited priors under a cross-model stability discipline — and seven findings. (1) The architecture's advantage over an audit-calibrated status quo (2.0–2.7× verified value per unit of budget) is invariant across behaviorally generated populations and peaks under low adoption; participation without a default layer is not weaker but non-functioning. (2) The informational quality of the planning vector dominates allocation quality, and distributed construction out-informs representative central construction everywhere its declared preconditions hold. (3) The share of citizens contributing explicit planning signals is structural (~2.5–5%) across all prior sources. (4) The deterrence stack is individually redundant and jointly indispensable: removing any single term costs almost nothing, removing all of them drops the architecture below the status quo — a finding that traveled back into the architecture's canon as a deployment-integrity rule. (5) The transition from the status quo to the full architecture is a dial, not a leap: a semi-open budget envelope yields monotone, near-linear gains from roughly a ten percent share upward. (6) Treating

the pipeline dynamically, the budget-release policy is a real institutional variable (pull-against-a-WIP-ceiling dominates calendar release), verification capacity is the pipeline's ceiling before it is the anti-fraud instrument, machine verification acts as capacity insurance whose lax accuracy requirement is *paid for* by the deterrence stack's slack, and adaptive adversaries neither erode nor invert the ranking while that stack is intact. (7) A real five-family model panel (600 ground-truth verdicts) measures machine verification directly: mechanical fraud detection is saturated, subtle fraud separates families, ensemble stacks run 2–6× worse than independence predicts, operating points are steerable by instruction and by evidence richness, the intact deterrence cascade covers even the worst measured verifier, and contraposed citizen evidence — independent producers with opposed interests — is the strongest single instrument measured, collapsing fraud by anticipated contradiction alone. Cross-framework docking and internal cross-engine replications support the results. Simulation evidence is not institutional proof; boundaries are declared throughout.

1. Introduction

The architecture paper asks whether today's technology permits a public-resource allocation design that improves on the incumbent institutional model, and answers with an architecture — Core v0 — validated by adversarial review and an eight-experiment simulation suite. Its governed object, and therefore ours, is the *discretionary, project-shaped portion* of a public budget (capital works, grants, local programs) — not payroll, transfers, or mandated services. That suite establishes the architecture's headline under its own assumptions. This paper attacks the assumptions.

Three families of question require a dedicated computational program. First, **behavioral realism**: the architecture paper imposes participation shares; what happens when participation *emerges* from an adoption process with friction, churn, social proof, and heterogeneous dispositions? Second, **mechanism accountability**: which of the architecture's mechanisms carry the results, which are redundant, and where are the cliffs — questions only ablation under adversarial pressure can answer. Third, **institutional dynamics**: the architecture specifies how money is allocated, disbursed, and verified, but real deployments must also decide *when* budget is released, *how much* verification capacity to fund, and *what happens over years* as adversaries adapt — variables the static suite cannot see.

Answering these requires methods worth reporting in their own right: engines whose every mechanism is anchored to a published architectural document (a traceability contract that makes "the simulation says" auditable), pre-registration with public prediction accounting in which refuted predictions are first-class findings, endogenous-participation generation, and a calibration path that uses large language models as *synthetic prior elicitors* — never as oracles — under a discipline that only cross-model-stable patterns inform anything.

Contributions: (i) the framework (Section 2); (ii) seven findings about the architecture (Sections 3–9), the last measured on a real five-model-family panel rather than simulated; (iii) cross-framework and cross-engine validation (Section 10); (iv) a demonstrated feedback loop between simulation and institutional design, in which findings traveled into the architecture's corpus through its adversarial pipeline as an operating-regime formalization and a deployment-integrity rule.

All results are simulation evidence about a model of the architecture — not institutional proof, not legal authorization, not pilot validation. Section 10 collects the declared boundaries.

2. Methods: the experiment framework

Engines. Four deterministic engines, all seeded (mulberry32 / Python `random`), version-stamped, and regression-anchored across versions: (A) a static adversarial architecture-comparison engine (JavaScript, dependency-free) implementing five institutional arrangements over a common project world; (B) a behavioral adoption model (Python; optional Mesa backend, byte-identical either way) generating populations — awareness, onboarding friction, activity, churn, role formation — under Core v0-conformant rules; (P) a planning-vector construction engine comparing how priority vectors are built; and (E) a longitudinal pipeline engine (v0.6–v0.8) adding budget-release policy, Poisson project arrivals, funding-window expiry, multi-cycle execution with milestone escrow, bounded verification capacity, executor reputation dynamics, machine-verification lanes, verification windows, and adaptive adversaries. Every result directory carries a metadata stamp (engine version, scenario, seed, runs); engine upgrades reproduce prior anchors exactly (the longitudinal anchor $V = 0.303844218531$ is bit-stable across v0.6→v0.8; the static engine's committed baselines are byte-identical across v0.4→v0.5.1).

The traceability contract. Every load-bearing mechanism maps to its anchor in the architecture corpus (the repository's traceability matrix): the public default rule, allocation secrecy, reputation-informs-never-excludes, evidence-triggers-never-certifies, conditional disbursement as the formal deterrence inequality, audit-anchored status-quo parameters, and the operating-regime vocabulary. Mechanisms without an anchor are either declared experiment-internal or listed as gaps. This converts conformance from a claim into a checkable artifact.

Adversarial comparison discipline. Architectures are compared on identical worlds (paired seeds; the same projects, opportunist draws, and citizen populations face every arrangement), under attacks (salience cascades, weak-verification shocks, fiscalizer collusion, agenda capture, coordinated signal bias), against a status quo calibrated to published audit-institution findings rather than to convenience — including the incumbent's actual control regime, not an opaque caricature. The primary metric is verified value per unit of budget (V): latent social value delivered, discounted by what independent verification can actually confirm.

A normative commitment, named. V presupposes that project-level social value exists prior to the mechanism and can be discovered and aggregated — the "latent social value" vector each world draws. That is a position, not a neutrality: the deliberative-constructivist tradition, in which value is *produced* by contestation rather than revealed by aggregation, is outside this model, as are redistribution, rights, and expressive politics that do not take the form of verifiable deliverables. Findings about "informational quality" of planning constructors are findings *within* this commitment; whether the architecture ranking survives an alternative objective (a min-share/Rawlsian target is the natural first test) is open and named as future work. Individual allocation privacy, a load-bearing mechanism referenced throughout, means ballot-style privacy of each citizen's directed allocation as an anti-coercion device — aggregates, budgets, projects, and every institutional act remain fully public.

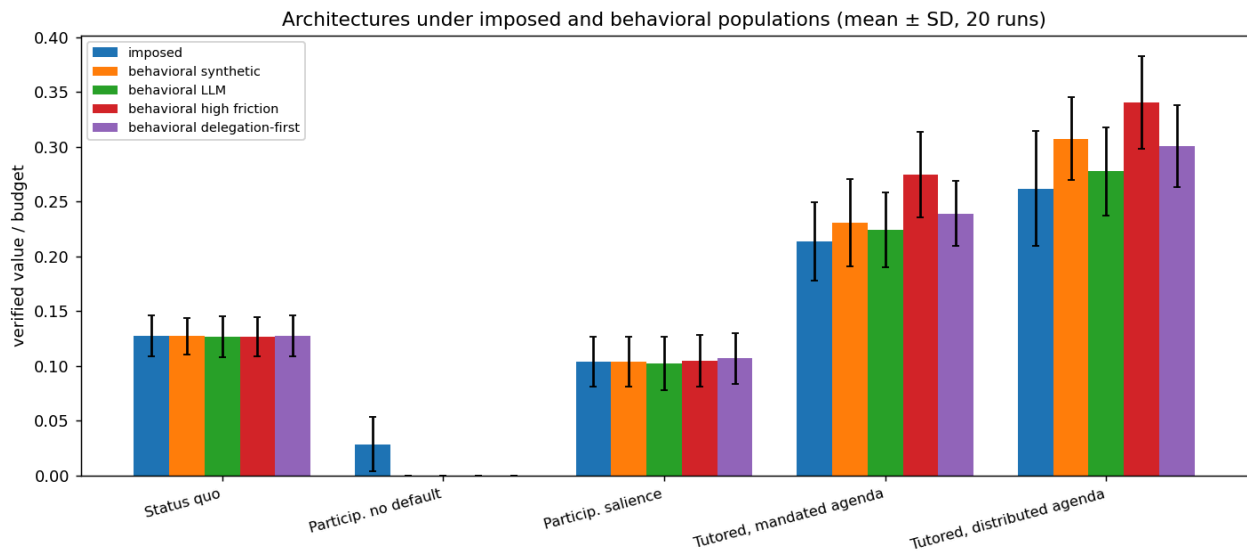
Endogenous participation (Experiment C). The behavioral model's emitted trajectories — activity shares, channel choices (direct, profile, delegation, default), planning attentiveness — replace the static engine's imposed participation blocks through a published mapping. The architecture comparison then reruns under populations the adoption process actually produces, including a high-friction and a delegation-first world.

LLM-elicited priors. Where behavioral parameters lack empirical data (dispositional willingness, channel preferences, delegation acceptance), we elicit distributions from language-model panels role-playing twenty population-weighted archetypes, under three rules: the instrument is identical to the human questionnaire that will eventually replace it; provenance is recorded as `llm-elicited`, a class that cannot be promoted to empirical; and only patterns stable across model families inform conclusions. The limits of that discipline are stated plainly: the cross-family check (a local open-weights family paired against a frontier hosted family) is exercised at $n=90$; the precision-bearing $N=1,000$ panel is single-family and by construction cannot detect its own fingerprint, so its "convergence by $n\approx 750$, composition drift ≤ 0.02 " is a within-model sampling statement, not calibration accuracy — a second frontier-scale family is named future work. Mean-sufficiency (using elicited means rather than full distributions downstream) was tested, not assumed: the committed distribution-versus-means analysis found downstream results insensitive to the collapse at these sample sizes, with the fixed-seed distributional runs as the check.

Pre-registration, with its verifiability stated honestly. Designs with named predictions were written before runs throughout, but the *git-verifiable* guarantee varies and we state it rather than blur it: the E-1a and E-1c designs were committed separately, minutes before their runs (independently checkable from history); the ablation and E-1b designs were committed atomically with their results (written first, but not separably provable); the semi-open quantification ran without a standalone design file and is classed as exploratory. A machine-readable predictions registry (`predictions.csv`: prediction $\rightarrow$ source $\rightarrow$ outcome) accompanies the repository. Of twenty-three registered predictions, eight were refuted or null in range and one materially transformed — each reported as a finding, several the most informative results of their phase.

3. Finding 1: the ranking is behavioral-population-invariant, and the default layer is existential

Under four behaviorally generated populations (synthetic baseline, LLM-calibrated, high-friction, delegation-first) plus the original imposed blocks, the ordering never changes: Core v0 with a distributed agenda delivers $2.05\text{--}2.69\times$ the status quo's verified value per unit of budget (paired 95% intervals exclude zero everywhere; between-base-seed spread of the ratio is $2.19\text{--}2.28\times$ on the calibrated population — statistical annex).



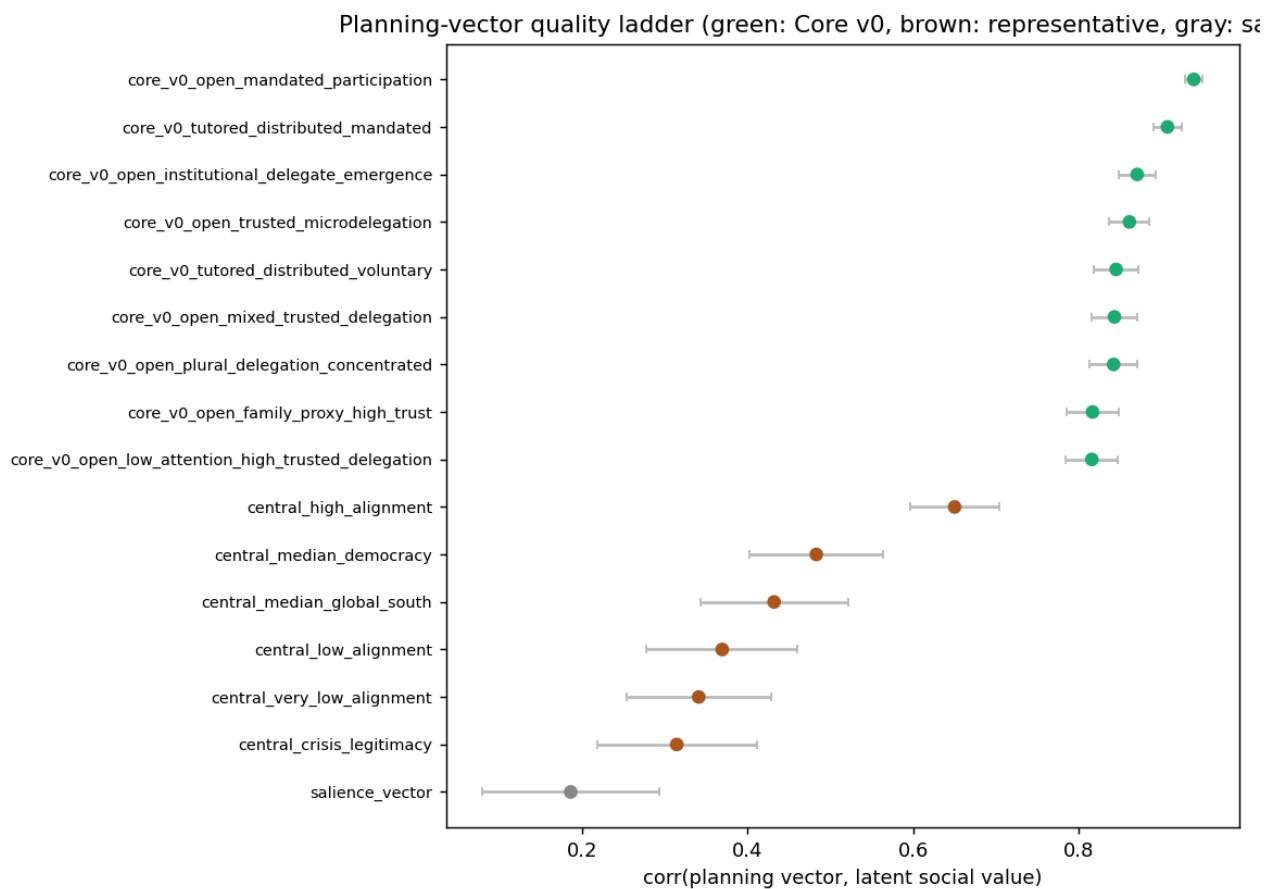
Verified value per unit of budget across the five architectures under imposed and behavioral populations (mean $\pm$ SD, 20 paired runs). The ordering is population-invariant; the no-default participatory variant collapses to zero under every behavioral population.

!Core-distributed to status-quo verified-value ratio per population. The advantage peaks under high friction — low adoption routed through a well-constructed default is the architecture's best case.](./figures/expc_core_advantage.png) Two structural results sit beneath the ratios. First, the advantage *peaks under low adoption* (high-friction: 2.69 $\times$): passivity routed through the default layer is the architecture's best case. The condition carrying that counterintuitive result must be named: it holds because the default vector is *good* — its informational quality is exogenous in these worlds (signal mix 0.66 under the honest-signal boundary) — and the ablation's sweep prices the dependence: degrading the mix to 0.2 attenuates the advantage to 1.63 $\times$ (never below it), while honest elicitation under gaming remains the architecture's declared open problem. Low adoption is the best case *given a well-constructed default*, not unconditionally. Second, the participatory variant without a default layer — the arrangement most civic-tech deployments implicitly build — collapses to **zero verified delivery under every realistic population**: with 3–6% active participation and no absorption rule, projects never reach funding thresholds. Participation without a default layer is not a weaker architecture; it is a non-functioning one. The design lesson is uncomfortable and quantified: the unglamorous default rule, not the participation feature, is what makes citizen allocation an institution rather than a poll.

4. Finding 2: the planning vector's informational quality dominates

The planning-vector program compares constructors on correlation with latent social value: the Core v0 channel family (attentive citizen signals + delegated signals, with voluntary or mandated participation) spans 0.815–0.939; representative central construction (coherent, bandwidth-limited, salience-guided planners across regime variants from crisis legitimacy to high alignment) spans 0.315–0.650; pure salience reaches 0.187. The families do not overlap: **the worst distributed channel out-informs the best representative regime** — but only where its declared preconditions hold (an aggregation institution exists; signals are honest and unbiased), and honest elicitation under gaming remains the architecture's declared open problem. Downstream, this gap is the dominant

term everywhere: the ablation's largest single-mechanism knockout is replacing the distributed vector with a central one ($\Delta V -0.062$), and the architecture-level sweep of vector informativeness moves the advantage from 1.63× to 2.35× — the largest lever any single parameter commands.



Planning-vector quality ladder: correlation with latent social value per constructor. The Core v0 channel family (top band) does not overlap the representative central family; salience-only sits at the bottom.

5. Finding 3: the attentive share is structural

Across adoption scenarios and prior sources alike, the share of citizens contributing explicit planning signals stays bounded in the low single digits: the committed range, 2.5–5.1%, is the spread *across adoption scenarios* (2.5% under high friction to 5.1% under AI-assisted onboarding), and every prior source — synthetic dispositions, the paired $n=90$ panels, the weighted $N=1,000$ panel — lands inside it. Attentiveness moves with adoption context within that band but never escapes it: a bounded disposition, not a scalable engagement outcome. Two epistemic notes before the consequences. The agreement between synthetic and LLM-elicited priors is **not** convergent validity — both are non-empirical sources with overlapping provenance — so within the program this result holds the status of a hypothesis pending the identical human instrument. Its external benchmark is instead the participatory-budgeting field record: three decades of PB deployments report active-participation rates in the low single digits of the eligible population (Porto Alegre's canonical years, the cross-national surveys collected by Sintomer and colleagues, and Wampler's Brazilian comparisons), the same band the model produces endogenously — while the simulated

polity omits what the field literature shows *moves* those rates: party mobilization, campaign cycles, and organized-group turnout, none of which are modeled beyond a scalar campaign-intensity parameter. What the finding then says is conditional and useful: designs that require more than single-digit spontaneous attentiveness are betting against both the model and the field record.

Three consequences. First, planning-layer designs must work at single-digit attentiveness — and do: the distributed vector's quality survives because delegation and defaults aggregate the attentive minority's information. Second, the architecture paper's imposed informed share (0.30 of participants) is behaviorally realizable: the LLM-calibrated adoption process produces 0.309 endogenously, and the architecture paper's own E8 experiment confirms the headline survives behavioral participation at $2.26\times$ [2.23, 2.30] versus $2.22\times$ imposed — both numbers recorded in the *master* repository's simulation-results (§E8), cited here as external evidence: E8 consumes this program's populations but runs, and is reproducible, in that repository, not this one. Third, the binding constraint on the control side is not adoption but the verification market — anticipating Finding 6.

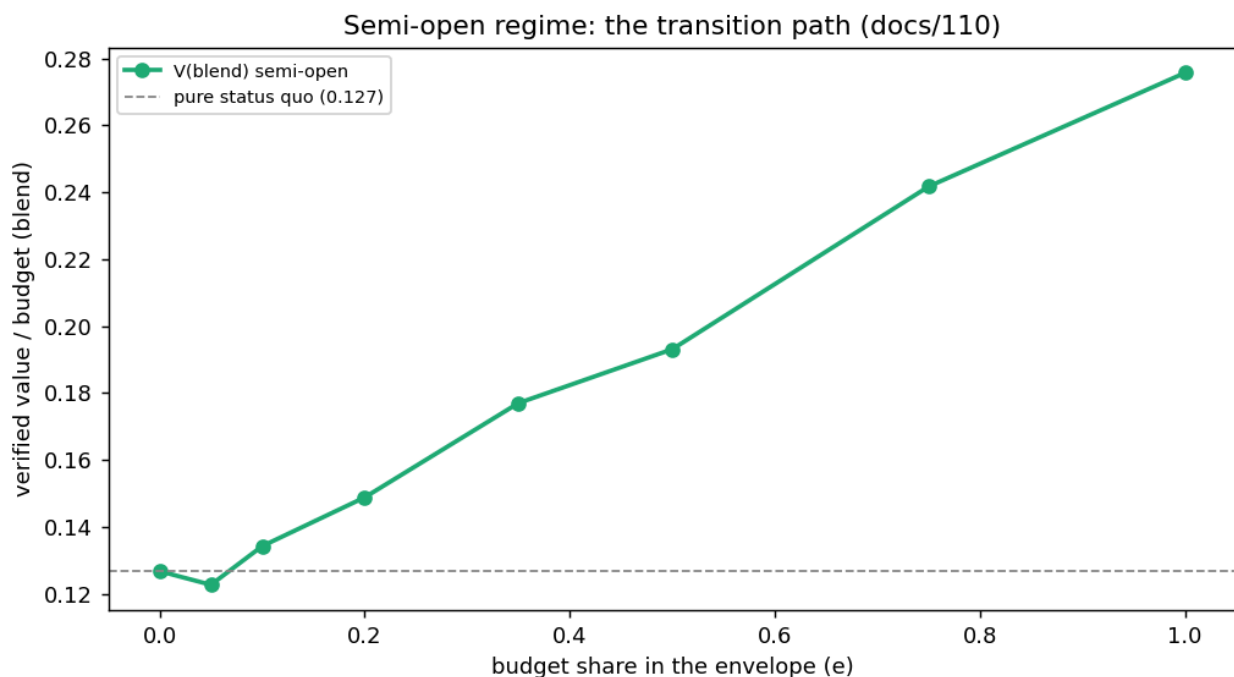
6. Finding 4: the deterrence stack — individually redundant, jointly indispensable

The pre-registered ablation knocked out each mechanism. Deterrence terms fall in a distinctive pattern: removing detection intensity, financial terms, or reputational memory *individually* costs $\leq 0.003 V$ — at the designed operating point the incentive-compatibility inequality holds with slack, so every single term looks worthless at the margin. Removing the **whole stack** collapses verified value by 60% and drops the architecture **below the status quo ($0.87\times$) with its selection quality intact**: distributed selection cannot compensate unverifiable delivery. The institutional threat is not an attacker but an implementer negotiating defensible concessions one at a time; the finding traveled through the architecture's adversarial pipeline (attack A041) and was resolved into its canon as a deployment-integrity rule — every scope publishes its deterrence-inequality margin, recomputed on every term change, with term reductions classed as material rule changes.

The attack program sharpened the picture. Fiscalizer collusion at plausible rates is *neutralized* by the stack's slack (the formal model's collusion-anticipation term made visible). Coordinated allocation bias among attentive citizens costs ≤ 0.007 at a fifty-percent coordinated share. The only attack that bites — capture of a published planning vector (-13% value, -35% selection quality at high severity) — presupposes a publishing choke point that exists only in the tutored-with-mandated-agenda regime; the distributed construction has no such surface, and its analog (signal corruption) is the robust case. The architecture's own regime doctrine absorbed this reading: capture pricing is an argument *for* distributed agenda construction, not against the architecture. Finally, the advantage *grows* with executor-pool dirtiness ($2.39\times$ at 70% opportunists): the dirtier the environment, the more the controls are worth.

7. Finding 5: the semi-open transition is a dial, not a leap

The architecture's operating-regime ladder defines a semi-open regime: a bounded budget envelope on protocol autopilot beside the authority's traditional budget. Its first quantification, as a fiscal blend over disjoint project pools: blended verified value rises monotonically and near-linearly with the envelope share — break-even near 8–10%, 1.52× at half the budget, 2.18× at full — with a single dip at five percent caused by two-project portfolio granularity, not by the regime. The envelope reaches full-architecture quality from roughly a 35% share; what grows thereafter is the share of budget enjoying it. For adoption politics this is the friendliest possible shape: no valley, no threshold, every increment pays, and the minimum sensible envelope is set by portfolio diversification, not by faith.



The semi-open transition path: blended verified value per budget as the envelope share sweeps 0–100%, against the status-quo reference. A dial, not a leap.

8. Finding 6: the pipeline view — release, verification, machines, and adversaries

The longitudinal program (Experiment E) makes time real: budget releases, projects arrive and expire, execution takes cycles under milestone escrow, verification has bounded capacity, reputation accumulates, adversaries adapt.

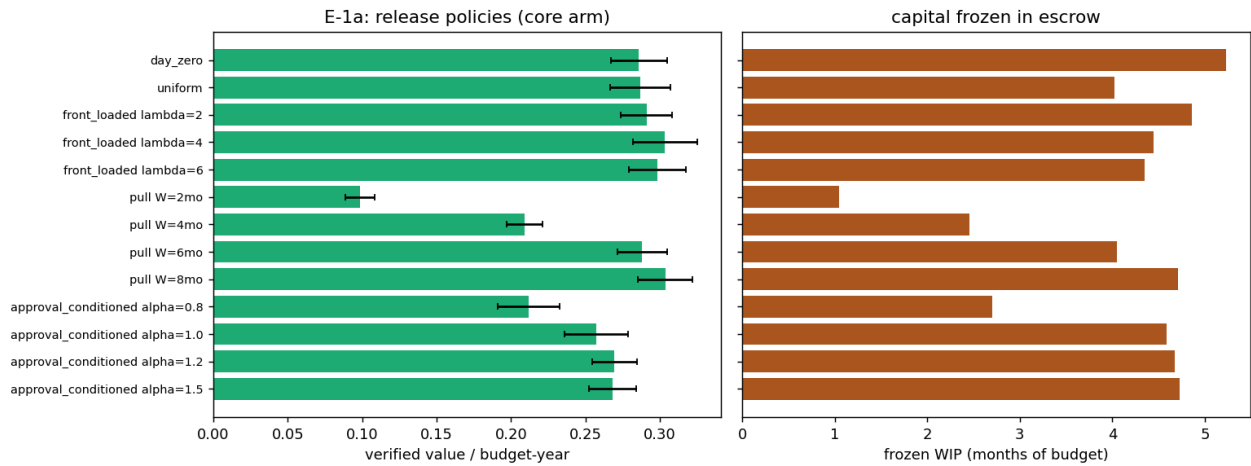
Release policy (E-1a). The authority's release schedule — a variable no architectural document regulated — matters institutionally. Day-zero release freezes 5.2 months of budget in execution escrow with the verification queue saturated; uniform monthly release (the implicit default of most public budgeting) maximizes project mortality through funding-window expiry; releasing against a **work-in-progress ceiling** (a pull policy) dominates the Pareto frontier at $W \approx 7 \text{ months of budget}$. The mechanism is Little's Law: *healthy WIP $\approx$ achievable throughput $\times$ cycle time*, and W tracks the *binding resource* — verification capacity when scarce, execution duration when

abundant. Feedback release conditioned on recent approval *flow* underperforms (0.269 vs 0.304); the correct conditioning variable is the outstanding-commitment *stock*, which is what pull implements. Under scarce verification capacity, no policy recovers the ceiling ($V \leq 0.255$) and over-release actively destroys value (0.236) — **verification capacity is the pipeline's ceiling before it is the anti-fraud instrument**.

A public-finance caveat belongs in the main text, not a footnote: the model's treasury does not lapse, while most appropriations regimes enforce fiscal-year annuality. The release rule is therefore stated *conditional on a carryforward instrument* — a capital fund, revolving fund, or multi-year investment program; the architecture's semi-open envelope is precisely such a vehicle — and under strict annuality it degenerates to within-year pull. An E-1a variant with lapsing funds and explicit encumbrance is pre-committed as the next engine run.

Machine verification (E-1c). Modeling an AI verifier as a triage system (auto-release for protocolizable evidence classes, referral-with-dossier, unconditional human sampling) with the deterrence coupling made explicit — sophisticated opportunists anticipate the end-to-end detection probability — yields three results. At scarce capacity, triage lifts the ceiling no release policy could ($0.254 \rightarrow 0.316$). The pre-registered accuracy crossover *never appears* up to a 20% false-pass rate: the dossier effect and the deterrence stack's slack absorb verifier error, so the accuracy requirement for a useful machine verifier is remarkably lax — **conditional on an intact stack**. And the human sampling lane's binding role is epistemic (measuring the machine's real error rate), not leak-plugging. Verification windows address a different failure: fiscalizer unresponsiveness costs up to 38% of delivered value, and a one-cycle timeout with automatic reassignment recovers ~80% of it; timeouts fix the tail, never the ceiling.

Adaptive adversaries (E-1b). Congestion-timed diversion under evidence staleness, evidence gaming against the machine lane up to skill 0.55, and eight years of dynamics all move verified value within noise on the intact configuration; the pull rule removes the congestion attack surface outright (backlog too short for evidence to stale), and even at engineered joint failure (scarce capacity + day-zero release, queue of thirty) the timing-aware adversary gains only fractions of leakage. On a **broken** stack, every verification arrangement — human, machine, gamed machine — lands within noise of the same collapse ($V \approx 0.114\text{--}0.118$): *verification is downstream of deterrence; nothing at the verification layer compensates a broken incentive floor*. Reputation compounding over eight years is null in range: the advantage is flat and robust from the first cycle, because a stack that deters from the start never dwells at the margin where memory would compound. The program's institutional closure fits in three lines: **size the capacity; meter the release; protect the margin**.



E-1a release policies: verified value per budget-year (left) and frozen work-in-progress in months of budget (right) across the policy grid. The pull family dominates the value-vs-freeze trade.

The verifier-displacement problem and its frontier (E-1d). A near-zero-price machine verifier is a *displacement threat* to the human fiscalizer market whose sampling is the machine's only audit: it collapses both the cost of fiscalization and the incentive to fiscalize, and with it the epistemic layer that measures the machine. The design answer, recorded in the architecture corpus and evaluated here, places humans permanently as the **second instance** — a mandated audit layer funded by a published control-budget floor (the machine's savings re-allocate to its own auditors; a complement by rule, not a competitor on price). The coverage frontier is **universal exposure, selective inspection**: every disbursement faces a nonzero published probability of human second-instance review, and the mandatory (probability-one) set comprises non-protocolizable evidence classes, stakes above a scope-published threshold, executors with thin track records, machine-flagged anomalies, complaint-triggered reviews, and post-model-change windows. The E-1d evaluation then corrected the design's own epistemics: with an intact deterrence stack, a verifier drifting from 5% to ~95% false-pass costs almost nothing ($V \sim -0.002$) and *passive sampling never detects it* — near-zero true positives make the false-pass rate unmeasurable at any affordable sampling rate. The healthier the system, the blinder its outcome audit. The frontier's epistemic instrument is therefore **seeded positive controls** — known-violation probes injected into the machine lane at rate q , giving drift-detection latency $\propto 1/(q \cdot \pi)$ independent of the deterred real diversion rate — while the random floor `s_min` serves exposure, appeals, and deterrence optics. Human second-instance review is thus *always on* as statistical exposure and seeded probing for every project, and *always total* only inside the mandatory set.

9. Finding 7: machine verification measured — a real five-family panel

The machine-verification results of Finding 6 used stipulated error rates; the F program replaced them with **measured** ones. Five model families — Qwen3.6-35B, Gemma 4-26B, and DeepSeek-R1-14B running locally; gpt-5.5 and Claude hosted — judged 120 ground-truth evidence vignettes committed before any run (600 verdicts, machine-scored), twice: once with mechanically-checkable

flaws, once with flaws requiring judgment (prices inflated against an in-bundle market reference, implausibly compressed timelines, coverage gaps, buried cross-reference contradictions, related parties, threshold structuring).

Mechanical fraud is saturated; subtle fraud separates the families. On mechanical flaws every family catches essentially everything (correlation untestable — declared). On subtle flaws:

family	false-pass (subtle fraud)	false-flag (clean work)
gpt-5.5	0.033	0.800
Claude	0.083	0.050
Gemma 4	0.119	0.458
DeepSeek-R1	0.217	0.317
Qwen3.6	0.000 (approves nothing)	1.000

The ensemble-independence illusion, measured. Pairwise false-pass correlations are positive (up to $\phi=0.43$); measured two-layer stacks run **1.9×–6.0× worse than the independence prediction**, with compressed timelines as the panel-wide blind spot. In the pipeline engine, a second layer pays only against independent-error fraud ($k^*=2$, benefit gone by $q=0.3$) and multiplies honest referrals.

Operating points are chosen, not given — by instruction and by evidence richness. A calibrated prompt moved gpt from 80% to 27% false-flags (at a recall cost, 97%→85%); richer evidence bundles moved Claude from 95% to 5% false-flags with the *same* prompt — objective comparison anchors turn paranoia into judgment, so evidence contracts should include the references (market benchmarks, duration bands, thresholds): judgeable-by-construction.

The cascade covers the measured failures (F-3). Feeding the measured operating points back into the pipeline: on an intact deterrence stack, even the worst measured verifier (21.7% false-pass) produces leakage indistinguishable from pure-human verification (0.0052 vs 0.0056), because the cascade removes the *attempts* upstream; the measured two-layer pair beats pure human outright (0.0020); and no verifier rescues a broken stack. Today's imperfect models suffice at the verification layer, provided the incentive cascade is intact — model progress is upside, not a prerequisite.

Contraposed citizen evidence is the strongest single instrument measured (F-4). Independent evidence producers with interests opposed to the executor — validated at ingestion, requiring no active dissent — collapse fraud by anticipated contradiction alone: leakage –57% at half coverage and –87% at full on the intact stack, and –72%/–95% on a *broken* stack — the only instrument in the program that substantially rescues a degraded cascade. Human review then concentrates where machines disagree, confidence is low, stakes are high, evidence contradicts, or dice say so: humans touch ~15–25% of projects lightly (dossier-assisted), roughly 70–85% less verification labor, while the *watching* stays distributed in the citizenry.

10. Cross-validation

Cross-framework dock. An independent Mesa-native Python reimplementations of the static architecture comparison (different language, paradigm, and RNG — Mesa 3.5.1 with Python's `random`, versus dependency-free JavaScript with `mulberry32`; equivalence assessed distributionally per Axtell et al.'s docking standard) reproduces the five-architecture comparison on the LLM-calibrated population at **15/15 PASS** (five architectures × three metrics, every $|\Delta_{\text{mean}}|$ within two pooled standard errors): verified value per budget 0.2780 vs 0.2774 for the distributed-agenda arm, exact zero-zero agreement on the no-default collapse, and matching leakage and selection-correlation distributions throughout (`docking/DOCKING_REPORT.md`). The largest residual divergence is disclosed rather than averaged away: the dock's status-quo cell lands at 0.1345 versus the reference's 0.1267 — inside the band ($|\Delta|$ 0.0077 vs 0.0114), but enough to move the dock-side headline ratio to $\approx 2.07\times$ versus the reference's $2.19\times$. The docking claim is therefore ordinal and distributional: same ranking, same collapse structure, ratios agreeing to within $\sim 5\%$ — not decimal-identical ratios. The equivalence criterion (two pooled standard errors per cell) was fixed in the docking brief before the port ran; a pre-specified TOST margin is the named upgrade for revision.

Internal cross-engine docks. Three replications across structurally different engines: the longitudinal engine's degraded-stack cell ($V = 0.114$, leak 0.057) reproduces the static ablation's joint-knockout cell (0.110, 0.053); the architecture paper's E8, consuming this program's behavioral populations in its own engine, reproduces its imposed-participation headline (2.26 vs 2.22); and the emergent attentive share (Experiment B) lands inside the range the planning-vector program had assumed independently.

Statistics. Paired-by-seed comparisons throughout; the annex reports 95% t-intervals — every headline paired difference excludes zero — plus the multiple-comparison statement: claims rest on pre-registered predictions, exploratory sweeps are labeled, and refuted predictions are published.

11. Limitations and declared boundaries

- **LLM priors are synthetic.** They carry model-family fingerprints; only cross-model-stable patterns informed conclusions, and the identical human instrument is the designed replacement. No `llm-elicited` quantity was promoted to empirical.
- **Honest elicitation under gaming is the architecture's declared open problem;** every engine assumes honest signals and says so. The planning-vector results hold conditional on that boundary.
- **Model boundaries:** no per-project authority veto stage (the tutored/semi-open decision-layer difference is unpriced); milestone objects are escrow schedules, not full evidence contracts; single-scope worlds (no federation); verification staleness parameters stipulated, directional claims only; the demand-smoothing instrument was implementation-biased and is demoted pending a delay-neutral scheduler (declared, not hidden).
- **Reputation-compounding null** is at this engine's project granularity; higher-throughput economies could compound faster.

- **Status-quo calibration** anchors to published audit-institution findings; jurisdictions differ, and the architecture paper's own calibrated-baseline rule governs what "status quo" means here. The incumbent's flatness across *participation* populations is by construction (it has no participation channel), and its institutional functions that V does not measure — electoral preference aggregation, deliberative conflict mediation, distributional bargaining, legitimacy production — are outside the metric and therefore outside every ratio reported here. Distributional-outcome metrics are the designed follow-up.
- **The legal layer is beneath the model, not inside it.** Citizen allocation directs funds to *projects*; executor award and payment run through existing procurement law (the architecture's allocation-act characterization keeps the citizen act pre-procedural), milestone-gated disbursement maps onto ordinary progress-payment practice, and statutory debarment is country law operating outside the platform — fully compatible with the platform itself never excluding. Procurement's cost, delay, protest processes, and set-aside obligations are unmodeled. Likewise, the architecture's fiscalizers are its own control market, *not* the constitutionally independent state audit bodies, which remain unchanged and appear here only as calibration sources; and the administrative-law status of machine-gated disbursement (due process, appealability — the complaint path is the designed appeal hook) is an open deployment question paired with the second-instance human design.
- **Scope of the governed object.** Most public spending is recurring services and transfers; the architecture governs the discretionary, project-shaped portion of a budget (capital works, grants, local programs). Every headline claim should be read at that scope, with the envelope share calibrated to a jurisdiction's real discretionary composition at deployment.
- **Simulation evidence, not institutional proof.** The program measures a model. Its role is to find breakpoints, rank mechanisms, and hand the architecture quantified questions — three of which it already answered into the architecture's canon.

12. Conclusions

Six findings survive an adversarial, pre-registered, behaviorally grounded program: the ranking is population-invariant and the default layer existential; planning-vector quality dominates and distributed construction wins where its preconditions hold; attentiveness is structural and single-digit; the deterrence stack is complements-not-substitutes with a published-margin rule as its deployment consequence; the transition is a dial; and the dynamic pipeline adds three institutional rules — size the capacity, meter the release, protect the margin — with machine verification as capacity insurance whose safety the deterrence stack itself underwrites, and with the human second instance as the machine's permanent auditor.

Two meta-results matter beyond this architecture. Methodologically, the combination — traceability contracts, pre-registration with public refutations, endogenous participation, LLM-prior calibration under cross-model discipline, and docking — is a portable recipe for computational institutional design. Institutionally, the program demonstrates a disciplined feedback loop between simulation and design: findings entered the architecture's corpus through its published attack–defense–resolution pipeline, with recorded verdicts. An honesty note belongs beside that claim: the

simulation, the architecture, the pipeline, and the verdicts are one author's ecosystem — the loop demonstrates *process discipline* (findings cannot silently change the design; refutations are published), not independent validation. The independence step is external: the architecture paper's prepared expert-review packets and, ultimately, the human calibration instrument and a bounded pilot. Within that boundary the claim stands: simulation did not flatter the architecture; it argued with it, refuted a third of its own predictions in public, and forced two design rules the architecture did not previously have.

Reproducibility statement

All engines, scenarios, results, figures (with accessible textual descriptions), designs, and prediction accounting are in the public repository, with one-command reproduction (`python tools/reproduce_all.py`) covering every committed artifact except the LLM panels, which are committed with full provenance and prompts. Code MIT, documents and results CC BY 4.0.

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